

Does the Filter Prefer the Past?

Measuring and Correcting Recency Bias in Confidence-Derived Evidence
Utility Signals for Retrieval-Augmented Alzheimer's Clinical Reasoning

MS Thesis Research Proposal

Candidate: _____

Supervisor: _____

Department: _____

Institution: _____

August 2026

Document status: research proposal. No experimental results are reported. Every quantitative statement concerning the proposed system is a hypothesis or a design target and is labelled as such.

Contents

Evidential Labelling Convention	4
Abstract	4
1. Introduction	5
1.1 Background	5
1.2 Problem Statement	5
1.3 Motivation and Clinical Significance	5
1.4 Research Gap	6
1.5 Aim and Objectives	6
1.6 Research Questions and Hypotheses	7
1.7 Scope and Delimitations	7
2. Related Work and Novelty Position	9
2.1 The Foundational System	9
2.2 Evidence Utility and Context Filtering	9
2.3 Knowledge Conflict, Confirmation Bias and Temporal Evaluation	9
2.4 A Directional Caution on Conflict-Aware Methods	9
2.5 Novelty Position	10
3. Proposed Solution: Overview	11
4. Methodology and Architecture	12
4.1 Notation	12
4.2 Component One: The Filter Recency-Bias Probe	12
4.3 Component Two: Three-State Currency	13
4.4 Component Three: The Contested State	13
4.5 Admission Score and Selection	14
4.6 Validity Controls	14
5. Implementation	16
5.1 Data and Corpus Construction	16
5.2 Evaluation Datasets and Provenance	17
5.3 Models, Hardware and Training	18
5.4 Minimum Viable Implementation	18
6. Experimental Design	19

6.1 Arms	19
6.2 Variables	19
6.3 Fairness Guarantees	19
6.4 Counterfactual Corpus-Update Procedure	20
7. Evaluation and Validation Strategy	21
7.1 Metrics	21
7.2 Decoupling Analysis	21
7.3 Statistical Analysis Plan	21
7.4 Ablations	22
7.5 Clinical Evaluation and Error Taxonomy	23
8. Expected Contributions and Outcomes	24
9. Timeline and Pre-Declared Pivots	24
10. Limitations and Ethical Considerations	25
References	26

Evidential Labelling Convention

Approximately one third of this design remains undetermined. The following tags appear throughout so that established evidence, authorial decisions, assumptions and open questions are never confused with one another.

Tag	Meaning
[PR]	Peer-reviewed or venue-accepted publication
[PP]	Preprint; not yet peer reviewed
[AU]	Authoritative non-academic source (Cochrane, guideline body, regulator)
[DES]	Design decision taken by this thesis
[ASM]	Assumption; not established by evidence
[HYP]	Falsifiable hypothesis to be tested
[OPEN]	Undetermined; requires specification before implementation

Abstract

Retrieval-augmented generation is widely adopted to mitigate hallucination and knowledge staleness in medical language models, yet the evidence-selection stage of such pipelines is rarely examined. The foundational system for this thesis, RAG-squared [PR], trains a small classifier to admit retrieved passages according to the perplexity reduction they induce in the base model — that is, according to how much a passage raises the model’s confidence.

This proposal argues that confidence gain is not equivalent to evidential support, and that confidence-derived utility signals may be systematically asymmetric with respect to evidence age, because passages consonant with a model’s pre-training-era priors raise confidence while passages carrying newer, dissonant guidance lower it. The claim is stated at the level of the *signal* rather than the artefact: the same information-theoretic formulation of evidence utility has been independently proposed as a general principle and now underlies reward signals in learned retrieval systems, so any bias would extend well beyond the specific filter examined here.

The primary contribution is a measurement instrument, the Filter Recency-Bias Probe, which quantifies admission asymmetry over matched temporal-counterfactual passage pairs while holding architecture, parameter count, training data, retrieval and reranking constant. The secondary contribution is a corrective admission policy, SCAF, that replaces confidence-derived utility with entailment-derived support, adds a three-state currency model (current, superseded, contested) and treats source authority as a tested variable rather than an assumed ordering.

Alzheimer’s disease is the evaluation domain because its diagnostic criteria and therapeutics changed substantially between 2023 and 2025, and because — as of April 2026 — one of those changes is under active scientific dispute, furnishing a natural test case for the contested state. Probe material is drawn from MedChangeQA [PR], an externally authored, peer-reviewed dataset, which eliminates the circularity that would arise from evaluating the system on items authored from the same documents it is designed to prefer.

1. Introduction

1.1 Background

Large language models perform strongly on medical question-answering benchmarks yet exhibit two persistent failure modes in clinical settings: hallucination, and reliance on knowledge frozen at pre-training time. Retrieval-augmented generation [PR] addresses both by conditioning generation on retrieved documents.

In the biomedical domain, however, retrieval introduces failure modes of its own. The largest expert evaluation of medical RAG conducted to date — eighteen clinicians and 80,502 annotations across 800 system outputs [PP] — reports that standard retrieval augmentation degrades medical performance under several conditions: only 22% of top-16 retrieved passages were judged relevant, 31% of queries returned no relevant passage at all within the top sixteen, and coverage of clinician-designated must-have statements reached only 33%. Retrieval quality, evidence selection and answer faithfulness are therefore distinct problems requiring distinct interventions.

1.2 Problem Statement

RAG-squared admits or rejects each retrieved passage using a classifier trained on labels derived from the perplexity differential $\Delta\text{PPL} = \text{PPL}(x) - \text{PPL}(x, d) \geq \tau$, with τ fixed at the top 25% of differentials. A passage is labelled helpful when conditioning on it reduces the base model's perplexity.

This criterion measures *belief shift*, not *evidential support* — a distinction that matters because belief shift is not neutral with respect to passage content. Two independently established findings bear on this. First, model adoption of retrieved evidence is confidence-mediated: the less confident a model is in its own prior, the more readily it adopts retrieved content [PP]. Second, models exhibit confirmation bias toward evidence partially consistent with parametric memory [PR].

The problem addressed here. If a filter's training signal rewards confidence gain, and confidence gain is elevated by passages consonant with the model's pre-training-era beliefs, the filter may be *directionally biased toward older evidence*. This has never been measured. Nor is it incidental to one system: the same information-theoretic formulation — utility as the shift a passage induces in the model's output distribution — has been independently proposed as a general principle [PP] and forms the basis of reward signals in learned retrieval.

1.3 Motivation and Clinical Significance

Alzheimer's disease is selected for three reasons, in ascending order of importance.

- **The evidence base moved recently and substantially.** The Alzheimer's Association revised criteria [PR] redefined the disease biologically, introduced Core 1 and Core 2 biomarker categories, and elevated accurate plasma biomarkers to diagnostic sufficiency. Appropriate-use recommendations for lecanemab [PR] and donanemab [PR] specify eligibility, mandatory APOE genotyping and ARIA monitoring schedules that did not previously exist.
- **Staleness here has direct clinical consequence.** A system reproducing pre-2024 consensus may recommend cerebrospinal-fluid or PET confirmation where a validated plasma assay now suffices, omit APOE-stratified ARIA risk counselling, or fail to identify anticoagulation as a contraindication.

- **Knowledge in this domain does not move monotonically forward.** In April 2026 a Cochrane systematic review [AU] of seventeen trials and 20,342 participants concluded that amyloid-beta-targeting monoclonal antibodies probably produce little to no clinically meaningful cognitive difference at eighteen months, against increased ARIA risk. The Alzheimer’s Association, the UK Dementia Research Institute and a Lancet commentary contested that conclusion within days.

There is consequently no single correct answer available to a system asked in 2026 whether a given patient should receive anti-amyloid therapy. Any architecture assuming that newer evidence supersedes older evidence would suppress one side of a live scientific controversy. This observation motivates the contested state specified in Section 4.4 and is, in the author’s assessment, the most clinically consequential element of the design.

1.4 Research Gap

Claim	Status	Evidence
Support-supervised filters outperform weaker filters in medical RAG	Established	[PP] arXiv:2511.06738 — relevance-supervised filter plus query reformulation, reported by two RAG-squared co-authors
Entailment-based assessment of context sufficiency	Established	[PR] Joren et al., ICLR 2025
Information-theoretic utility filtering	Established	FILCO [PP]; formalised as information gain [PP]
Lightweight distilled evidence critic with risk-aware abstention	Established	[PP] CoRM-RAG, arXiv:2605.01302
Temporal benchmarks for outdated medical knowledge	Established	[PR] Vladika et al., Findings of EMNLP 2025 — MedRevQA and MedChangeQA
Filter-stage admission asymmetry with respect to evidence age	Open	No located work. Existing studies measure confirmation bias at the generator, or temporal awareness parametrically — not at the admission stage
Explicit representation of contested evidence in medical RAG	Open	[PR] Javadi et al., ECIR 2026 call for contradiction-aware filtering but implement none

Gap statement. The literature establishes that confidence-based utility signals exist, that they are widely used, and that generators exhibit confirmation bias. It does not establish whether the *selection stage itself* is temporally asymmetric, nor does it provide any mechanism for representing evidential disagreement. These two gaps define this thesis.

1.5 Aim and Objectives

Aim. To determine whether confidence-derived evidence utility signals exhibit a measurable recency bias in medical retrieval-augmented generation, and to specify and evaluate a minimal corrective admission policy for Alzheimer’s clinical reasoning.

1. Reproduce the RAG-squared baseline, including retraining its filtering model, since the trained checkpoint is not distributed.
2. Construct a date-annotated, Alzheimer’s-scoped retrieval corpus with per-passage temporal metadata.
3. Construct FRB-PAIRS from MedChangeQA, with an accompanying permutation control set.

4. Implement and execute the Filter Recency-Bias Probe across two label functions and two model backbones.
5. Specify and implement SCAF: entailment-derived support, three-state currency, tested authority weighting, and an abstention and disagreement gate.
6. Evaluate SCAF against RAG-squared and the current state-of-the-art medical filtering pipeline.
7. Conduct a blinded clinical evaluation on open-ended Alzheimer’s vignettes and a structured error analysis.

1.6 Research Questions and Hypotheses

ID	Research question	Hypothesis (direction only)
RQ1	Do confidence-derived utility signals admit passages asymmetrically with respect to age, holding content, tier and length constant?	H1: $\Delta(\text{perplexity}) > 0$. H2: $\Delta(\text{perplexity}) > \Delta(\text{support})$.
RQ2	Does any asymmetry replicate across filter backbones — is it a property of the signal or of one model?	H3: asymmetry replicates with the same sign.
RQ3	Does an entailment-derived label reduce unsupported and superseded claims?	H5: unsupported-claim rate falls.
RQ4	Is any reduction attributable to admission policy rather than mediated by retrieval quality?	H6: a direct arm effect persists after conditioning on recall.
RQ5	Does an explicit contested state improve clinical appropriateness on unresolved questions?	Assessed by blinded expert rubric.
RQ6	Is accuracy preserved on time-invariant medical QA?	H7: non-inferiority within a pre-registered margin δ .

A further hypothesis functions as a **validity gate rather than a finding**. **H4:** on the permutation control, $\Delta = 0$. Should the control fail, H1 to H3 are uninterpretable and must not be reported as results. [OPEN] The non-inferiority margin δ must be pre-registered with explicit justification; two accuracy points is offered for discussion, not adopted.

1.7 Scope and Delimitations

In scope: the evidence admission stage of a single-pass RAG pipeline; Alzheimer’s disease and related dementias; English-language public corpora; filter models of at most one billion parameters; open-weight generators of at most eight billion parameters, plus one optional commercial comparison.

Out of scope, with justification:

- *Retriever improvement.* The retriever is deliberately frozen — the central internal-validity guarantee (Section 6.3). This is defensible on evidence: [PP] arXiv:2511.06738 tested BM25, MedCPT and Qwen3-Embedding-0.6B and found no retriever consistently superior, concluding that retriever replacement alone cannot overcome the intrinsic limitations of the architecture.
- *Agentic and reinforcement-learned retrieval.* The dominant 2026 direction, but out of scope on a single-GPU budget. One non-competing agentic reference point is retained for context.
- *Private clinical data.* No institutionally approved clinical notes are assumed; all resources are public.

- *Clinical validation.* The expert evaluation demonstrates mechanism at small scale and will not be described as clinical validation.

2. Related Work and Novelty Position

2.1 The Foundational System

RAG-squared [PR] comprises three components. *Rationale-based query formulation* replaces the question with a chain-of-thought rationale generated by the base model, the original query being excluded because concatenation exceeds the retriever's maximum length. *Balanced retrieval* draws equal passage quotas from PubMed, PMC full text, clinical practice guidelines and eighteen textbooks — 37.6 million documents, 116.7 million passages, 564.2 GB — using MedCPT [PR] for dense retrieval and cross-encoder reranking. *Rationale-guided filtering* uses a Flan-T5-large classifier (770M parameters) trained on perplexity-differential labels. Reported gains are +6.1, +3.8 and +0.9 average accuracy points for Llama-3-8B-Instruct, Meerkat-7B and GPT-4o across MedQA, MedMCQA and MMLU-Med.

Two properties are load-bearing here. First, the paper motivates itself with hallucination and outdated knowledge but **measures neither**; all headline results are multiple-choice accuracy, and the only open-ended evaluation comprises twenty-five queries scored with ROUGE-L and BERTScore. Second, the system contains **no temporal representation of any kind** — no publication date, guideline version or supersession relation appears in the corpus, index, retriever, reranker or filter.

A third property, noted in independent analysis of the base paper and adopted here, is that **perplexity labels are base-model-specific**: the filter must be retrained per backbone, and transferability is unverified. This has direct cost consequences (Section 5.3).

2.2 Evidence Utility and Context Filtering

FILCO [PP] introduced context filtering by lexical and information-theoretic measures, including conditional cross-mutual information — the direct ancestor of the perplexity differential. Sufficient Context [PR] formalised entailment-based assessment of whether retrieved context suffices to answer a query. Most directly relevant, a 2026 formulation [PP] defines utility as information gain — the divergence between the model's output distribution with and without the evidence — stating that a passage has high utility precisely when it induces a significant shift in the model's belief relative to its parametric knowledge. This is the general form of the criterion this thesis examines, and its independent proposal confirms that the signal family is expanding rather than obsolescent.

2.3 Knowledge Conflict, Confirmation Bias and Temporal Evaluation

ClashEval [PP] quantifies the tension between parametric prior and retrieved evidence, establishing confidence-mediated adoption. Xie et al. [PR] establish confirmation bias toward partially consonant evidence. Both operate at the *generator*. Vladika et al. [PR] establish that all eight evaluated models exhibit memorization of outdated medical knowledge, and contribute MedRevQA (16,501 pairs from systematic-review abstracts, PubMed 2000 to January 2024) and MedChangeQA (512 pairs whose verdict changed over time). Javadi et al. [PR] evaluate medical RAG under publication-year stratification and explicitly call for contradiction-aware filtering without implementing it.

None of this work examines the admission stage. That is the gap this thesis occupies.

2.4 A Directional Caution on Conflict-Aware Methods

CARE [PR] is frequently proposed as a template for conflict handling in retrieval pipelines. Its stated purpose, however, is resolving “the conflict between incorrect external context and correct parametric knowledge” — training a context assessor to discern unreliable context and steer toward the more reliable source, which in its framing is frequently the model’s own parameters.

A temporal system requires the **inverse**: overriding outdated parametric knowledge in favour of newer external evidence. Importing this logic unmodified would reinforce precisely the staleness such a system exists to correct. [DES] CARE’s context-assessor architecture is adaptable and is cited here as prior art, but the directional inversion is explicit in this design and must not be left implicit.

2.5 Novelty Position

Not claimed as novel: entailment-based context selection; information-theoretic utility filtering; lightweight distilled critics with abstention; selective prediction; temporal medical benchmarks; Alzheimer’s-specific RAG.

Claimed as novel, with the boundary stated: filter-stage temporal asymmetry as a measured property, and the contested state as an implemented output condition in a medical retrieval pipeline.

Principal novelty threat, stated plainly. CoRM-RAG [PP] is substantially closer to this work than a surface reading suggests. Its abstract states that maximising relevance “paradoxically promotes the retrieval of sycophantic evidence that reinforces hallucinations”, a failure it terms the relevance-robustness gap; it addresses confirmation bias directly and distils a lightweight evidence critic performing risk-aware abstention. It therefore shares the premise, the diagnosis and the architecture. **The differentiator is the perturbation axis and the object of measurement:** CoRM-RAG perturbs *queries* with simulated cognitive biases to build adversarial robustness; this thesis perturbs *evidence age*, holding query and content constant, to measure a directional property of the admission signal itself. That distinction is real but will not be granted unless argued explicitly.

[ASM] **Urgency.** The information-gain formulation and CoRM-RAG both appeared in May 2026. Connecting either to temporal evidence is a foreseeable next step for other groups; the contribution window is estimated at twelve to eighteen months. This assumption drives the sequencing in Section 9, which front-loads the probe.

3. Proposed Solution: Overview

Three principles govern every subsequent decision. **Isolate one variable:** every stage upstream of admission is frozen and byte-identical across arms, so any observed difference is attributable to admission policy alone. **Measure before intervening:** an intervention that improves a metric without establishing a mechanism is engineering; establishing the mechanism first makes the intervention explanatory. **Design for an informative null:** the experiment must be equally publishable whether the hypothesis holds or fails.

The proposed system therefore comprises a **diagnostic instrument** (the probe, Section 4.2) and a **corrective policy** (SCAF, Sections 4.3 to 4.5), evaluated within an experimental design whose fairness guarantees are specified in Section 6.3. Figure 1 presents the complete architecture.

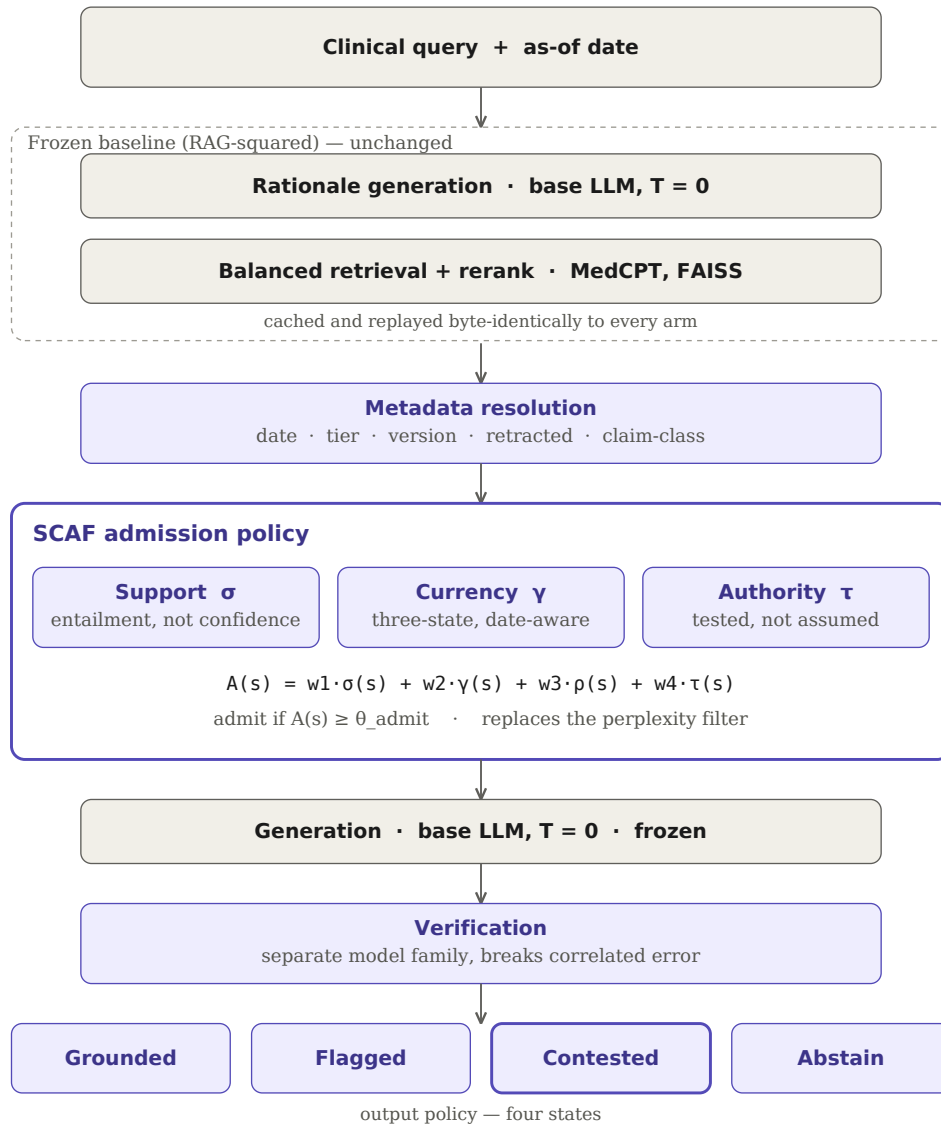


Figure 1. Revised end-to-end architecture. Grey components are inherited from the base paper and held byte-identical across all experimental arms; violet components are proposed. The perplexity filter is replaced, not supplemented; the three thin additions — metadata resolution, verification and output policy — exist to make the contribution measurable and the output behaviour safe.

4. Methodology and Architecture

4.1 Notation

Symbol	Definition	Symbol	Definition
q, r	Query; generated rationale	$\sigma(s)$	Entailment-derived support score
t_q	Query as-of date	$\gamma(s, q)$	Currency score
$s, m(s)$	Candidate passage; its metadata	$\rho(s)$	Rank-normalised reranker score
$K(s)$	Claim-classes assigned to s	$\tau(s)$	Source-authority weight
T	Supersession table	$A(s)$	Admission score
$\psi(q)$	Time-sensitivity indicator	S^*	Admitted passage set

4.2 Component One: The Filter Recency-Bias Probe

Pair construction. A temporal-counterfactual pair consists of two passages addressing the same claim on opposite sides of a known change point, matched on source tier, on token length within a tolerance band [OPEN], and on topical similarity to the query [OPEN]. [ASM] Matching is necessary because older and newer statements of the same claim plausibly differ in hedging convention, citation density and prose style, any of which a filter may respond to instead of age. Matching reduces but cannot eliminate this; Section 4.6 supplies the control that detects it.

Measurement. For a filter f , the admission asymmetry is the expectation over pairs of $P(\text{admit} \mid \text{older}) - P(\text{admit} \mid \text{newer})$. A value greater than zero indicates preferential admission of older evidence.

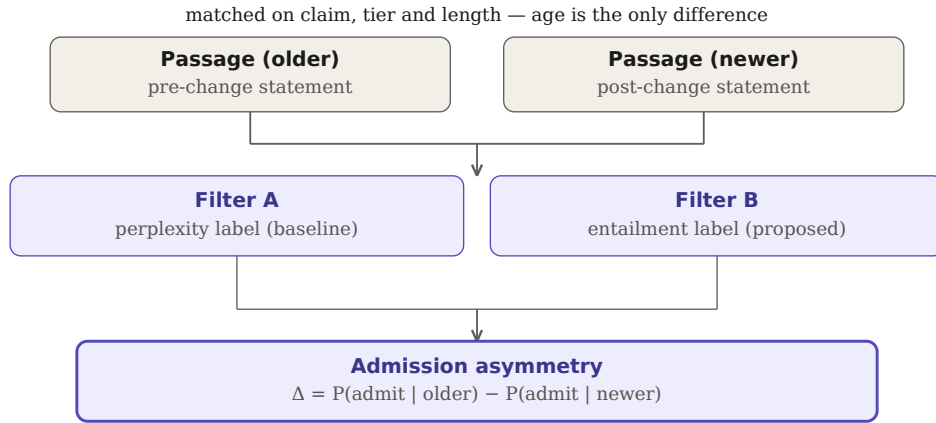


Figure 2. Probe design. The two filters share backbone, parameter count, training data and hyperparameters; only the label function differs. This identity is what licenses attributing the measured asymmetry to the signal rather than to capacity, architecture or retrieval.

Label functions. The baseline label reproduces the base paper: a passage is helpful if it flips the model from incorrect to correct, ambiguous cases resolved by the perplexity differential at the top 25% threshold, remaining cases discarded. The proposed label is derived from entailment against the verbalised gold answer at training time:

$$h^* = \text{verbalise}(Q, \text{gold answer})$$

```

sigma* = P_entail(premise = s, hypothesis = h*)

y(s)  = 1      if sigma* >= theta_hi
        0      if sigma* <= theta_lo
        discard otherwise      # mirrors the base paper's discard branch

```

[DES] Both filters are trained on general medical QA and applied zero-shot to Alzheimer’s items, mirroring the base paper’s own cross-dataset transfer claim. This removes any suspicion that domain gains derive from domain-specific fine-tuning, and keeps the evaluation sets uncontaminated.

Inference-time scoring. The gold answer being unavailable at inference, $\sigma(s)$ is the maximum entailment probability over the hypothesis set — verbalised answer options for multiple-choice items, or up to eight atomic claims decomposed from the rationale for open-ended items. A discriminativeness diagnostic (the margin between the highest and second-highest entailment) is recorded; a passage with high support but low margin entails several options equally and is decision-uninformative. [OPEN] Whether this diagnostic enters the admission score.

4.3 Component Two: Three-State Currency

```

gamma(s, q, t_q) =

    0                      if retracted(s) or withdrawn(s)
    1                      if psi(q) = 0
    delta * 2^(-(t_q - date(s))/H)  if superseded(s, t_q)
    2^(-(t_q - date(s))/H)          otherwise

```

Two design decisions require justification.

Why hard rejection is confined to retraction and formal withdrawal. Claim-class tags are produced by an automatic matcher with only a sampled manual audit. [ASM] If tagging precision falls materially below unity, a hard supersession gate silently and irreversibly deletes correctly relevant evidence at a rate proportional to the tagging error. Retraction and withdrawal, by contrast, are objective and externally verifiable. Supersession therefore down-weights rather than deletes.

Why the decay is conditioned on $\psi(q)$. An unconditional recency prior would penalise correct older sources on time-invariant questions — APOE biology, neuropathological staging, cholinesterase pharmacology — trading a currency gain for an accuracy loss. Conditioning is what makes the non-inferiority hypothesis plausible at all. [OPEN] Whether $\psi(q)$ is implemented as a rule over the claim-class taxonomy or as a trained classifier; evaluation sets carry manual labels regardless.

4.4 Component Three: The Contested State

A claim-class is contested when two passages of that class carry opposing conclusions, both dated within a contest window [OPEN] and both above a minimum source tier. When triggered, the policy **must admit at least one representative of each position**, and the output emits the contested state with both dates and both source identities.

[DES] **The contested test is evaluated before the supersession test.** Reversing the order would classify the April 2026 Cochrane review as merely superseding the appropriate-use recommendations, presenting a live controversy as resolved. The ordering is therefore load-bearing, not incidental.

[ASM] The principal technical risk is distinguishing genuine contradiction from apparent contradiction arising from scope difference: “effective in early-stage disease” against “not effective in moderate disease” is a scope qualifier, not a conflict, and a naive detector will conflate them. This is acknowledged as a limitation rather than solved, and is why the contested state is positioned as a secondary contribution demonstrating mechanism rather than a validated capability.

4.5 Admission Score and Selection

$$A(s) = w_1 * \sigma(s) + w_2 * \gamma(s, q) + w_3 * \rho(s) + w_4 * \tau(s)$$

[DES] The reranker score uses rank-based rather than min-max normalisation, so that $A(s)$ is comparable across queries and a global admission threshold is well defined. [DES] Source authority $\tau(s)$ is included but treated as a **tested variable**: excluding it would allow a recent preprint to outrank a current guideline, while assuming a fixed ordering — guidelines above systematic reviews, for instance — encodes a position that the April 2026 Cochrane episode shows to be contestable. The ordering is therefore an ablation (A12), not a premise. [OPEN] Weight selection uses a reduced parameterisation; a full simplex grid is computationally infeasible.

Algorithm 1: SCAF-ADMIT

Input: S_{rr} (reranked candidates with ρ), q , r , t_q , $\psi(q)$, T

Output: S^* (ordered admitted set), gate state

```

1  H <- build_hypotheses(q, r)
2  for s in S_rr:
3      sigma(s) <- max over h in H of P_entail(s, h)
4      gamma(s) <- currency(s, t_q, psi(q), T)
5  Cand <- { s in S_rr : gamma(s) > 0 }      # retraction / withdrawal only
6  for s in Cand: A(s) <- w1*sigma + w2*gamma + w3*rho + w4*tau
7  Cand <- { s in Cand : A(s) >= theta_admit }
8  if contested(c, t_q) for any class c present in Cand:
9      S* <- one representative of each position, then fill by A(s)
10     return S*, CONTESTED
11 S* <- top-k_final by A(s)
12 if S* empty or max sigma(s) < theta_support: return S*, ABSTAIN
13 Supp <- { s in S* : sigma(s) >= theta_support }
14 if max gamma(s) over Supp < theta_stale: return S*, FLAGGED
15 return S*, GROUNDED

```

Complexity is bounded by the entailment scoring: at most 32 candidates by 8 hypotheses, so 256 short forward passes on a sub-billion-parameter model per query, plus scalar operations. This is negligible relative to generation.

4.6 Validity Controls

Control	Mechanism	Failure consequence
V1 Permutation control	FRB-PAIRS duplicated with publication dates randomly reassigned; Δ must collapse to zero	Blocking. If the control fails, the probe is responding to prose-era features rather than recency and the primary result is uninterpretable
V2 Backbone replication	The label comparison repeated on a second, modern filter backbone	The claim narrows from a property of the signal to a property of one model

Control	Mechanism	Failure consequence
V3 Cached candidate replay	Upstream stages execute once; the candidate set is serialised and replayed byte-identically	Differences between arms would no longer be attributable to admission policy
V4 Decoupled verifier	The post-hoc claim verifier uses a different model family from the admission scorer	Correlated error: a claim the filter mis-scores would be mis-verified identically, making verification a mirror rather than a check
V5 Negative control set	Alzheimer's claims that did not change across the study window	A currency effect could not be distinguished from a general preference for recent, better-formatted documents

5. Implementation

5.1 Data and Corpus Construction

The full base-paper corpus (564.2 GB) is not tractable on the assumed hardware. The corpus is therefore domain-scoped, and the identical scoped corpus is used by every experimental arm, making scoping a controlled constant rather than a confound.

Corpus	Source	Scoping rule	Estimated scale [OPEN]
PubMed-AD	PubMed baseline via E-utilities	MeSH: Alzheimer Disease, Dementia, Cognitive Dysfunction, Amyloid beta-Peptides, tau Proteins, Lewy Body Disease, Frontotemporal Dementia	$2-4 \times 10^5$ abstracts
PMC-AD	PMC Open Access subset	Same filter, open-licence only	$2-4 \times 10^4$ full texts
CPG-AD	Clinical practice guideline collection	Dementia-relevant guidance	$\sim 10^3$ documents
Textbooks	Established medical textbook set	Neurology, psychiatry, internal medicine	18 volumes
Currency pack	2024 revised criteria; lecanemab and donanemab recommendations; ARIA guidance; Cochrane CD016297	Manual ingestion	20-40 documents

Metadata schema. Each passage carries: chunk and document identifiers; source tier; publication date; journal and persistent identifier; guideline family, version and supersession pointer; retraction and withdrawal flags; claim-classes; section header; character span.

Preprocessing. Sliding-window chunking at 256 tokens with 32-token overlap [DES], sized against the article encoder’s 512-token limit with headroom for a prepended title and section header. Exact deduplication by content hash; near-duplicate detection by MinHash over five-gram shingles at Jaccard 0.85, retaining the newest and highest-tier cluster member. [DES] This matters specifically here: preprint-to-journal duplicates would otherwise allow an old claim to survive under a new date. Guideline passages respect section boundaries, because a recommendation must never be separated from its qualifying conditions — ARIA monitoring cadence severed from the APOE stratification it applies to, for instance.

Date propagation and its risk. Every passage inherits its parent document’s publication date. [ASM] Guideline PDFs are expected to yield poorer date coverage than structured XML; uncorrected, this would systematically penalise the guideline corpus — the opposite of intent. Date imputation from issue metadata is applied and the per-corpus null rate is a reported figure.

Claim-class tagging. A hand-built taxonomy of forty to eighty classes covering biomarker criteria, staging, plasma biomarkers, amyloid imaging, eligibility, ARIA monitoring, APOE, pharmacotherapy, behavioural management, deprescribing and differential diagnosis. Assignment by keyword-and-embedding matching with manual verification of a 300-passage random sample, reporting precision and recall. [ASM] This is the weakest link in the pipeline, and is precisely why Section 4.3 confines its consequences to down-weighting rather than deletion.

5.2 Evaluation Datasets and Provenance

[DES] **Provenance separation is mandatory and is the single most important correction in this revision.** An earlier formulation authored evaluation items, the supersession table and the currency corpus from the same twenty-odd documents, rendering a positive result unfalsifiable by construction. The probe’s primary material is now drawn from MedChangeQA [PR] — externally authored, peer-reviewed in provenance, publicly released, and independent of any supersession table this thesis builds.

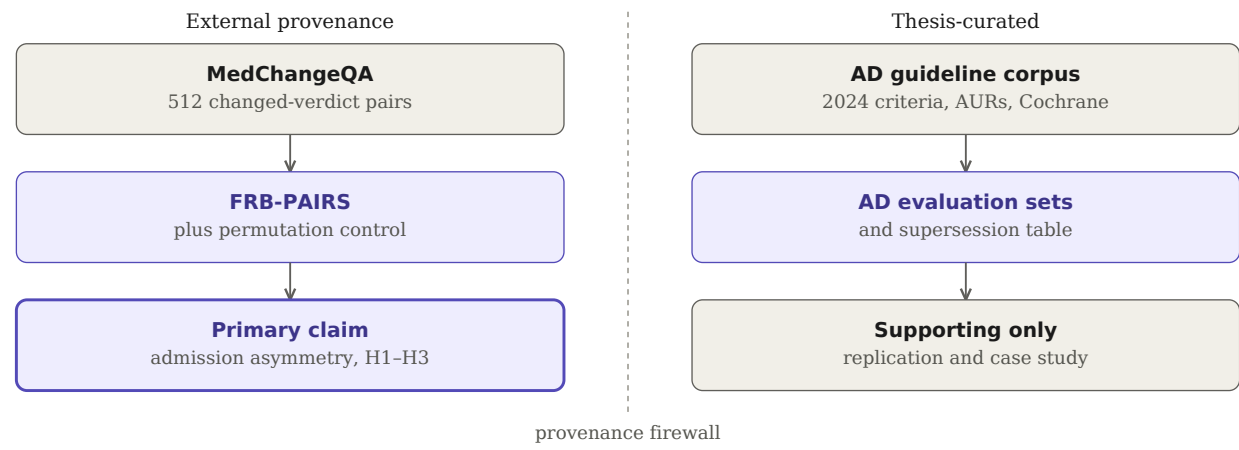


Figure 3. Evaluation provenance. The primary claim rests on the externally sourced lane; thesis-curated material supports replication and case study only. The lanes must not cross.

Set	Target size	Provenance	Purpose
FRB-PAIRS	150–300 pairs	Derived from MedChangeQA [PR]	Primary: admission asymmetry
FRB-PERMUTED	Same pairs	Dates randomly reassigned	Validity gate (H4)
NEG-CONTROL	~80 items	Claims stable across the window	Detects spurious recency preference
CONTESTED-AD	~40 items	Anti-amyloid efficacy before and after CD016297	Tests the contested state
ADCUR-QA	~120 items	Authored from documents <i>excluded</i> from the supersession table	Domain replication
AD-VIGNETTE	60–80 items	Open-ended clinical vignettes	Blinded expert evaluation
MedQA, MedMCQA, MMLU-Med	As published	Existing [PR]	Non-inferiority only
MedXpertQA	As published	Existing	Unsaturated accuracy comparison

[ASM] MedQA, MedMCQA and MMLU-Med approach saturation for strong models — GPT-4o is reported at 0.922 on MedQA [PP] — and are therefore retained solely for non-inferiority testing rather than as headline metrics. MedRevQA is sourced from PubMed up to January 2024 and structurally cannot contain post-June-2024 criteria items, which is why the domain replication set remains necessary but is no longer load-bearing.

5.3 Models, Hardware and Training

Role	Selection	Rationale
Retriever	MedCPT query and article encoders	Baseline fidelity; remains competitive — a 2025 alternative exceeds it by 0.0012 average accuracy on the standard medical RAG benchmark [PP]
Reranker	MedCPT cross-encoder	Baseline fidelity
Filter, primary	Flan-T5-large (770M)	Identical to the base paper; this identity licenses the label ablation
Filter, replication	Modern sub-billion encoder [OPEN]	Distinguishes a property of the signal from a property of one model
Entailment teacher	DeBERTa-v3-large fine-tuned on public biomedical NLI [OPEN]	Credentialed clinical NLI corpora are treated as optional and never load-bearing
Generator, primary	Llama-3-8B-Instruct	Largest reported baseline gain; 8K context; fits 24 GB
Generator, secondary	Meerkat-7B	Medically tuned, short context — tests sensitivity to context budget

Hardware [ASM]. One 24 GB GPU; at least eight CPU cores and 64 GB system memory, index construction and deduplication being memory-bound; one to two terabytes of solid-state storage. Domain scoping is what makes this sufficient.

Training cost, and the principal budget risk. The currency gate and output policy require no training. The entailment teacher and each filter student require hours. The dominant cost is regenerating the baseline labels: the base paper’s filter checkpoint is not distributed, and label generation requires one base-model forward pass per question-passage pair. Over the full training split this approaches millions of passes — and because **perplexity labels are base-model-specific, the exercise repeats per backbone**, giving four filter trainings rather than two. [DES] A pre-declared subsample of twenty to thirty thousand items is proposed, with cost verified empirically on one hundred items before the sample size is fixed. This is treated as a blocking item.

5.4 Minimum Viable Implementation

[DES] The following constitutes a complete and defensible thesis should all later phases fail: a date-annotated domain-scoped index restricted to abstracts and the guideline pack; the baseline reproduced on a 200-item subset with subsampled labels; FRB-PAIRS and the permutation control constructed from MedChangeQA; the label-function comparison executed; and H1 and H4 reported with confidence intervals. This requires no supersession table, no taxonomy at scale, no expert evaluation and no full-text parsing — deliberately, so that the deterministic and inexpensive half of the work is reachable independently of the expensive and logistically risky half.

6. Experimental Design

6.1 Arms

Arm	Description	Admission policy
B0	Closed-book reference	None
B1	Retrieval and rerank, no filter	None
B2	Foundational baseline (base-paper reproduction)	Flan-T5-large, perplexity labels
B3	Current state of the art — support-supervised filtering with query reformulation [PP]	Relevance-supervised filter
B4	Agentic reference point — iterative follow-up retrieval	Reported for context, not as a competitor
P	Proposed — SCAF	Support, three-state currency, tested authority, gate

[DES] B3 is required rather than optional. Comparing only against the base paper would benchmark the thesis against a system the field has already improved upon — and improved upon by two of that paper’s own authors.

6.2 Variables

Independent: admission policy; filter label function; filter backbone; currency gate state; authority weighting; corpus snapshot; generator; decay half-life, supersession discount, thresholds and context budget. **Dependent:** admission asymmetry; accuracy; unsupported and contradicted claim rates; citation precision and recall; retrieval recall, precision, mean reciprocal rank and normalised discounted cumulative gain; context relevance; redundancy; evidence coverage; outdated-evidence and superseded-answer rates; update-flip rate; abstention rate and calibration; cost per correct answer; expert rubric scores. **Controlled:** corpus and index build; chunking; retriever and reranker checkpoints and parameters; candidate depth; prompts and decoding; random seeds; hardware.

6.3 Fairness Guarantees

1. **Upstream identity.** The reranked candidate set is computed once, serialised and replayed byte-identically. Arms are distinct functions over one cached candidate list.
2. **Matched context budget.** All arms receive at most five passages. A matched-length variant, padding the proposed system with the next-best rejected passage, is reported as a sanity check.
3. **Matched filter capacity.** The baseline and proposed filters share backbone, size, training data volume and hyperparameters; only the label function differs.
4. **Prompt parity.** The main comparison uses one template without date annotations, so gains are attributable to which passages were admitted rather than to date cues. Date-annotated prompting is evaluated separately as a co-primary condition, because the staleness warning is the clinical payload of the system.
5. **No test-set tuning.** All thresholds and weights are selected on validation data and frozen before any test run, against a selection objective declared in advance.
6. **Contamination control.** Filters are trained only on general medical training splits; evaluation sets are finalised after training data are frozen.

6.4 Counterfactual Corpus-Update Procedure

[DES] This procedure isolates the pipeline’s knowledge-update responsiveness from parametric knowledge without retraining anything. An index is built containing only documents dated before the June 2024 criteria revision, and a second containing all documents. Identical models and prompts are executed against each, and the *update-flip rate* is reported: the proportion of items whose answer correctly transitions from the superseded to the current gold. Because generator weights are frozen and only the corpus changes, any difference is a property of the pipeline rather than of the model.

7. Evaluation and Validation Strategy

Standing commitment. Retrieval improvement is not treated as evidence of hallucination reduction. The two are measured on separate instruments and their relationship is tested explicitly rather than assumed.

7.1 Metrics

Target problem	Metrics	Known weakness
Recency bias (primary)	Admission asymmetry Δ with paired bootstrap; permutation-control Δ	Depends on the quality of pair matching; V1 is the detector
Hallucination	Unsupported and contradicted claim rates; citation precision and recall; entity-attribution error rate	Automatic judges are imperfect — see the validation guard below
Retrieval relevance	Recall@k, precision@5, mean reciprocal rank, nDCG@10 against pooled human judgements; context relevance; redundancy; evidence coverage	Pooled judgement is annotation-expensive
Currency	Outdated-evidence rate; temporal correctness; superseded-answer rate; guideline-version correctness; update-flip rate	Depends on supersession-table completeness
Safety behaviour	Abstention rate; appropriate against over-abstention; risk-coverage area; coverage at fixed accuracy	Trade-off surface, not a scalar win
Efficiency	Cost per correct answer; end-to-end latency	—

[DES] **Judge validation guard.** Automatic claim labels are validated against human labels on a stratified sample with Cohen’s kappa reported. [OPEN] The sample must be sized by a precision target rather than chosen for convenience. Should kappa fall below 0.6, automatic rates are demoted to secondary and expert ratings become primary. This guard is declared in advance.

[DES] **Mandatory dual reporting.** Every rate metric is reported twice — conditional on answering, and with abstentions counted as failures. A system that abstains more computes its rates over a smaller, easier denominator; a result presented only the first way is not interpretable.

7.2 Decoupling Analysis

To test rather than assume the retrieval-to-hallucination link: per-item rank correlation between change in retrieval recall and change in unsupported-claim rate, with confidence interval; a mixed-effects mediation model reporting the arm coefficient with and without recall as mediator, and the proportion mediated with a bootstrap interval; and the subset of items where the admitted set differs between arms while recall is unchanged — any difference there cannot be a retrieval effect.

7.3 Statistical Analysis Plan

Hypothesis	Test	Unit of analysis
H1, H2, H3	Paired bootstrap, 10,000 resamples; mean asymmetry with 95% interval	Matched pair

Hypothesis	Test	Unit of analysis
H4 (validity gate)	Equivalence test against a pre-specified band	Matched pair
H5	Mixed-effects logistic regression; arm fixed, random intercepts for question and claim within question	Atomic claim
H6	Mixed-effects mediation; arm coefficient with and without mediator	Item
H7	One-sided non-inferiority test on paired proportions against δ	Item
Currency metrics	McNemar's exact test on paired outcomes; Cohen's h	Item
Expert ratings	Mixed-effects cumulative-link regression; Cliff's delta	Rating

[DES] Claims are clustered within questions, so a naive per-claim chi-squared test would be anti-conservative; hence the mixed-effects specification for H5. Multiplicity is controlled by Holm-Bonferroni across the primary family at a family-wise level of 0.05, with secondary hypotheses reported uncorrected and labelled as such.

[DES] **Pre-registration.** Hypotheses, the non-inferiority margin, significance level, tests, minimum effect of interest and the pivot trigger are written and timestamped *before any contact with test data*. [OPEN] A power analysis determines evaluation-set sizes; sizes follow from a pre-specified minimum effect of interest rather than the reverse. Should achievable power prove low, that is reported as a limitation rather than concealed by post-hoc adjustment of the target effect. No expected numerical values appear anywhere in this proposal.

7.4 Ablations

ID	Configuration	Question answered
A1	Perplexity against entailment label, backbone and data identical	Primary causal claim
A2	Permutation control	Validity gate
A3	Second backbone replication	Signal against artefact
A4	No filter	Does any filter help?
A5	Support only, currency disabled	Contribution of currency
A6	Prompt with and without date annotation	Admission effect against date cue
A7	Currency only, support excluded	Contribution of support
A8	Contested state removed	Contribution of disagreement handling
A9	Abstention gate removed	Contribution of abstention
A10	Teacher at inference against distilled student	Distillation cost
A11	Sweep of half-life, discount, thresholds and context budget	Sensitivity
A12	Source-authority ordering varied and removed	Recency against authority trade-off
A13	Hard supersession gate restored	Cost of the safety-motivated softening
A14	Generator transfer	Backbone generality

A1 to A3 are load-bearing and are scheduled first.

7.5 Clinical Evaluation and Error Taxonomy

Three evaluators are targeted: one specialist with dementia experience, plus two of a senior trainee, clinical pharmacist or final-year medical student. [ASM] Recruitment is the largest schedule risk; the design remains valid with two evaluators and a reduced item set, reported as a limitation. Outputs from four arms are presented with arm identity removed and order randomised per item under a sealed key generated before packet assembly. Five ordinal dimensions with written anchors — factual correctness, evidence support, currency, clinical appropriateness and safety, and completeness — are supplemented by two binary items. Krippendorff's alpha is reported *before* arm differences; dimensions below 0.4 are reported descriptively only.

Every failure is assigned by a fixed decision procedure over logged intermediate state: retrieval failure; reranking failure; **filter over-rejection**; filter under-rejection; temporal failure; **evidence insufficiency**; hallucination; contradiction; clinical reasoning error; abstention error. [DES] Over-rejection and evidence insufficiency are rarely reported in the literature and are measurable here precisely because the candidate set is cached and the corpus enumerable. Separating retrieval failure from evidence insufficiency, and hallucination from reasoning error, is the analytical return on the logging design.

8. Expected Contributions and Outcomes

C1 (primary). The first controlled measurement of admission asymmetry with respect to evidence age in a medical retrieval pipeline, with architecture, capacity, training data and retrieval held constant, replicated across two backbones and validated by a permutation control.

C2. SCAF: an admission policy replacing confidence-derived utility with entailment-derived support, adding a three-state currency model with an explicit contested condition, and treating source authority as a tested variable.

C3. Released artefacts: FRB-PAIRS with its permutation control, the domain replication set, the supersession table, and a timestamped pre-registered analysis plan.

Expected to hold. Non-inferiority on time-invariant items, since the support signal should not degrade performance where recency is irrelevant; and some positive asymmetry for the perplexity-labelled filter, motivated by the confidence-mediated adoption and confirmation-bias findings in Section 2.3.

Expected to disappoint, and pre-declared as such. The accuracy arm on saturated benchmarks will show little movement; these are retained for non-inferiority only, and this is stated before results rather than after. Automatic hallucination metrics will be fragile: zero-shot filtering F1 in the 0.44 to 0.52 range is reported in the literature [PP], rising to 0.62 with fine-tuning, and a judge operating in that range cannot support fine-grained claims.

Principal risk to the expected result [ASM]. With 150 to 300 matched pairs, a modest asymmetry may fail to clear a conventional threshold. The most likely adverse outcome is not a clean negative but an *inconclusive* one, which is materially worse for a thesis. Pre-registration and adequate pair count are the only defences, which is why the power analysis precedes pair construction.

9. Timeline and Pre-Declared Pivots

Phase	Weeks	Deliverable	Gate
0. Power analysis and pre-registration	1–2	Timestamped analysis plan	Filed before any test-set contact
1. Corpus construction with dates	3–8	Scoped index, metadata store, null-date report	Date coverage and per-corpus null rates reported
2. Baseline reproduction (B2, B3)	6–12	Baseline retrained; state of the art standing up	Within one to two points on the domain subset
3. FRB-PAIRS and controls	10–16	Pairs from MedChangeQA, permutation set, negative control	Tier and length matching verified
4. Probe execution (A1–A3)	14–20	Asymmetry per configuration	Pivot decision taken here, in month five
5. SCAF implementation	20–28	Three-state currency, contested state, gate	End-to-end run without degraded states
6. Evaluation	26–34	All metrics across arms; judge validation	Kappa threshold or expert ratings become primary
7. Expert evaluation	32–38	Blinded ratings, agreement, adjudication	Krippendorff’s alpha per dimension

Phase	Weeks	Deliverables	Gate
8. Error analysis and write-up	36-44	Taxonomy distribution, decoupling analysis, thesis	Full citation verification completed

[DES] Phase 4 precedes Phase 5 deliberately. If the primary claim fails, that must be discovered in month five, when a pivot is inexpensive, rather than in month nine.

Trigger	Pre-declared response
A1 shows no label effect	Headline shifts to the currency and contested contributions; H1 and H2 reported as null results, which remain publishable given the strength of the prior expectation
A2 permutation control fails	Primary result withheld; pair construction revisited before any further analysis
Proposed system regresses beyond 6 on general benchmarks	Support-only becomes the default; currency becomes an opt-in gate; the trade-off surface is reported
Clinician recruitment fails	Structured checklist evaluation against clinician-authored answer keys, reported explicitly as a weaker instrument

10. Limitations and Ethical Considerations

Methodological limitations. The supersession table is small, manual, domain-scoped and non-exhaustive. The contested detector cannot reliably distinguish genuine contradiction from scope difference; it demonstrates mechanism, not validated capability. Absolute performance figures are not comparable to published baseline results because the corpus is domain-scoped — only within-study arm contrasts are interpretable. Findings are demonstrated for one disease area; generalisation to other rapidly updating domains is a hypothesis rather than a result.

Acknowledged design tension. This thesis studies the admission stage while the field’s own expert data indicate that retrieval is the dominant bottleneck — 22% top-16 relevance and 31% of queries with no relevant passage. Freezing retrieval buys causal attribution at the cost of a lower achievable ceiling and a reduced effective sample. [DES] This trade-off is deliberate and is stated explicitly rather than left for an examiner to raise.

Ethics. All corpora are public and openly licensed; no patient-identifiable data are used; credentialed resources are optional and never load-bearing. Clinician evaluators participate as expert raters rather than as research subjects. [OPEN] Institutional review requirements for expert rating studies must be confirmed before recruitment. The system is a research prototype: no output will be used for clinical decision-making, and the expert evaluation will not be described as clinical validation. [DES] In reporting the anti-amyloid case, the thesis will present the Cochrane conclusion and the responses of the Alzheimer’s Association, the UK Dementia Research Institute and Lancet commentators as an ongoing scientific disagreement, and will not adjudicate between them.

References

Grouped by evidential status. Where authorship could not be independently verified, the work is cited by title and identifier rather than by an inferred author list.

Peer-reviewed and venue-accepted [PR]

- [1] Choi, E., Park, J., Lee, H., & Lee, J. (2025). Conflict-Aware Soft Prompting for Retrieval-Augmented Generation. *Proceedings of EMNLP 2025*, 26981–26995. DOI 10.18653/v1/2025.emnlp-main.1371. arXiv:2508.15253.
- [2] Chung, H. W., Hou, L., Longpre, S., et al. (2024). Scaling instruction-finetuned language models. *Journal of Machine Learning Research*, 25(70), 1–53.
- [3] Cummings, J., Apostolova, L., Rabinovici, G. D., et al. (2023). Lecanemab: Appropriate Use Recommendations. *Journal of Prevention of Alzheimer’s Disease*, 10(3), 362–377. DOI 10.14283/jpad.2023.30.
- [4] Hendrycks, D., Burns, C., Basart, S., et al. (2021). Measuring massive multitask language understanding. *ICLR 2021*.
- [5] Jack, C. R., Jr., Andrews, J. S., Beach, T. G., et al. (2024). Revised criteria for diagnosis and staging of Alzheimer’s disease: Alzheimer’s Association Workgroup. *Alzheimer’s & Dementia*, 20(8), 5143–5169. DOI 10.1002/alz.13859.
- [6] Javadi, S., Mirabi, A., Gangar, N., & Ofoghi, B. (2026). Contradictions in Context: Challenges for Retrieval-Augmented Generation in Healthcare. *ECIR 2026*, Springer LNCS 16483. DOI 10.1007/978-3-032-21289-4_3.
- [7] Jeong, S., Baek, J., Cho, S., Hwang, S. J., & Park, J. C. (2024). Adaptive-RAG. *NAACL 2024*, 7029–7043.
- [8] Jin, D., Pan, E., Oufattole, N., et al. (2021). What disease does this patient have? *Applied Sciences*, 11(14), 6421.
- [9] Jin, Q., Kim, W., Chen, Q., et al. (2023). MedCPT: Contrastive pre-trained transformers with large-scale PubMed search logs. *Bioinformatics*, 39(11), btad651.
- [10] Joren, H., et al. (2025). Sufficient Context: A New Lens on Retrieval Augmented Generation Systems. *ICLR 2025*. arXiv:2411.06037.
- [11] Kwon, W., Li, Z., Zhuang, S., et al. (2023). Efficient memory management for LLM serving with PagedAttention. *SOSP 2023*, 611–626.
- [12] Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*, 33, 9459–9474.
- [13] Li et al. (2024). DALK: Dynamic Co-Augmentation of LLMs and Knowledge Graphs to Answer Alzheimer’s Disease Questions. *Findings of EMNLP 2024*. arXiv:2405.04819. [Full author list to be confirmed.]
- [14] Pal, A., Umapathi, L. K., & Sankarasubbu, M. (2022). MedMCQA. *CHIL 2022*, 248–260.
- [15] Rabinovici, G. D., Selkoe, D. J., Schindler, S. E., et al. (2025). Donanemab: Appropriate Use Recommendations. *Journal of Prevention of Alzheimer’s Disease*, 12(5), 100150. DOI 10.1016/j.tjpad.2025.100150.
- [16] Sohn, J., Park, Y., Yoon, C., Park, S., Hwang, H., Sung, M., Kim, H., & Kang, J. (2025). Rationale-Guided Retrieval Augmented Generation for Medical Question Answering. *NAACL 2025*, 12739–12753. DOI 10.18653/v1/2025.naacl-long.635. arXiv:2411.00300.
- [17] Vladika, J., Dhaini, M., & Matthes, F. (2025). Facts Fade Fast: Evaluating Memorization of Outdated Medical Knowledge in Large Language Models. *Findings of the ACL: EMNLP 2025*, 9161–9174. DOI 10.18653/v1/2025.findings-emnlp.487. arXiv:2509.04304.
- [18] Xie, J., Zhang, K., Chen, J., Lou, R., & Su, Y. (2024). Adaptive Chameleon or Stubborn Sloth: Revealing the behavior of large language models in knowledge conflicts. *ICLR 2024*. arXiv:2305.13300.

- [19] Xiong, G., Jin, Q., Lu, Z., & Zhang, A. (2024). Benchmarking Retrieval-Augmented Generation for Medicine. *Findings of ACL 2024*, 6233–6251. arXiv:2402.13178.

Authoritative non-academic [AU]

- [20] Nonino, F., Minozzi, S., Sambati, L., Del Giovane, C., Baldin, E., Bassi, M. C., De Santis, C., Gonzalez-Lorenzo, M., Vignatelli, L., Filippini, G., & Richard, E. (2026). Amyloid-beta-targeting monoclonal antibodies for people with mild cognitive impairment or mild dementia due to Alzheimer’s disease. *Cochrane Database of Systematic Reviews*, Issue 4, Art. No. CD016297. DOI 10.1002/14651858.CD016297.

Preprints, not peer reviewed [PP]

- [21] Kim, H., Sohn, J., Gilson, A., et al. (2025). Rethinking Retrieval-Augmented Generation for Medicine: A Large-Scale, Systematic Expert Evaluation and Practical Insights. arXiv:2511.06738.
- [22] Wang, Z., Araki, J., Jiang, Z., Parvez, M. R., & Neubig, G. (2023). Learning to Filter Context for Retrieval-Augmented Generation (FILCO). arXiv:2311.08377.
- [23] Wu, K., Wu, E., & Zou, J. (2024). ClashEval: Quantifying the tug-of-war between an LLM’s internal prior and external evidence. arXiv:2404.10198. [Publication venue not independently verified.]
- [24] Beyond Semantic Relevance: Counterfactual Risk Minimization for Robust Retrieval-Augmented Generation (CoRM-RAG) (2026). arXiv:2605.01302. [Authorship and venue not independently verified.]
- [25] Large Language Models Lack Temporal Awareness of Medical Knowledge (TempoMed-Bench) (2026). arXiv:2605.13045.
- [26] Utility-Oriented Visual Evidence Selection for Multimodal Retrieval-Augmented Generation (2026). arXiv:2605.13277.
- [27] LLM-Specific Utility: A New Perspective for Retrieval-Augmented Generation (2025). arXiv:2510.11358.
- [28] Enhancing Biomedical RAG through an Advanced Re-Ranking Retriever (2025). arXiv:2510.04757.
- [29] R2MED: A Benchmark for Reasoning-Driven Medical Retrieval (2025). arXiv:2505.14558.
- [30] Agentic-R: Learning to Retrieve for Agentic Search (2026). arXiv:2601.11888.
- [31] Provence: Efficient and Robust Context Pruning for Retrieval-Augmented Generation (2025). arXiv:2501.16214.

Citation verification note

Independently verified during preparation of this proposal: the base paper’s pagination, DOI and technical details; Vladika et al. (dataset sizes and PubMed sourcing window); Choi et al. (venue, pagination and DOI); Joren et al. (lead authorship); Javadi et al. (venue, DOI); Cochrane CD016297 (authorship, issue and publication date); and Kim, Sohn, Gilson et al. (by full-text retrieval, including the annotation counts and retrieval statistics cited in Section 1.1).

The following appeared in earlier working documents for this project and must be confirmed against the ACL Anthology, DOI resolution or publisher record before appearing in the thesis. They are deliberately absent from the reference list above: META-RAG; Med-R2; MedHallu; ADRD-Bench; Deceptive Grounding; FaithMed; MedAbstain; AD-CARE; MedCite; MedGraphRAG; AMG-RAG; T-GRAG; CARE-AD; the DETeCD-ADRD guideline citations; and the DALK item breakdown.